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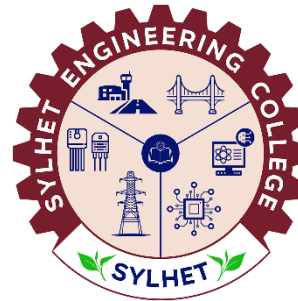
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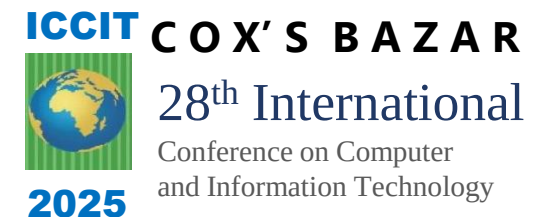
BEYOND SENTIMENT: THE INFLUENCE OF GAME PRICING ON PLAYER FEEDBACK IN STEAM REVIEWS

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Presentation Outline

- Introduction & Motivation
- Literature Review
- Problem Statement
- Methodology
- Results & Analysis
- Key Findings
- Implications
- Conclusion & Future Work

Introduction

- **Gap:** Limited research on how game pricing influences player sentiment and review topics.
- **Hypothesis:** Price may not directly affect sentiment, but fundamentally shapes player expectations.
- **Challenge:** 730,945 reviews from 743 games on Steam—how to understand player feedback?
- **Relevance:** Critical for developers and publishers making pricing decisions.

Research Gap

Sentiment Analysis

- Previous works focuses on sentiment analysis.
- There exist few researches on unsupervised topic analysis for specific area.

Research gap

- Topic analysis in a broader area.
- Correlation between price and user review content.

Literature overview

Previous works

Bian et al. (2021) examined reliability of Steam reviews and reported 15% of 33,450 reviews as spam

Yu et al.(2021) Employed Latent Dirichlet Allocation on 1.2 million esports reviews.

Li et al(2021) Used topic modeling in data driven framework to assess game playability

Guzsvinecz (2025): Analyzed 418,483 reviews of Souls like titles using LDA to uncover recurring theme

IM Urrzia et al (2021): Aspect based sentiment analysis of user created game reviews

Research Questions

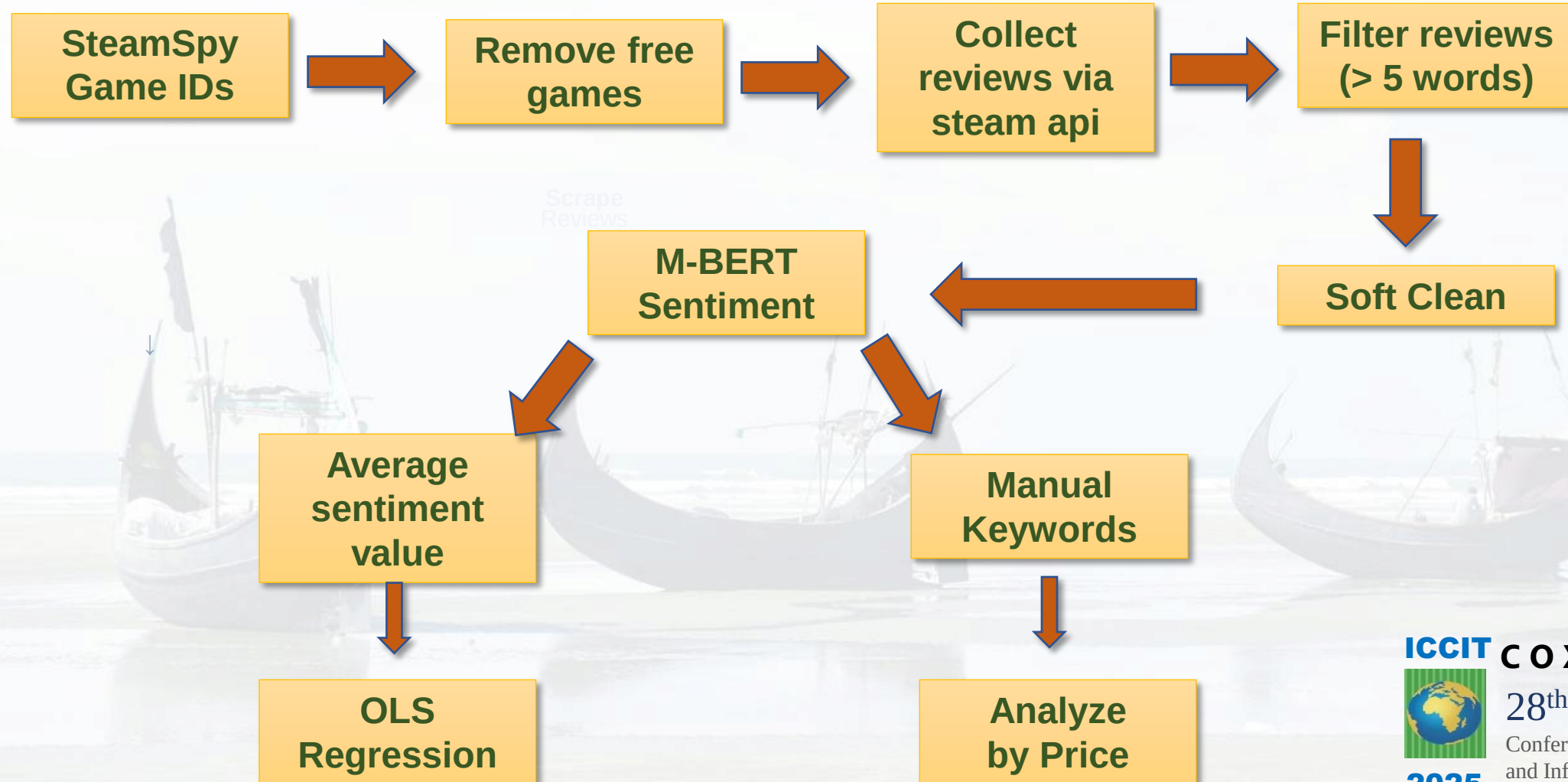
RQ1

Does game pricing affect user sentiment expressed in reviews?

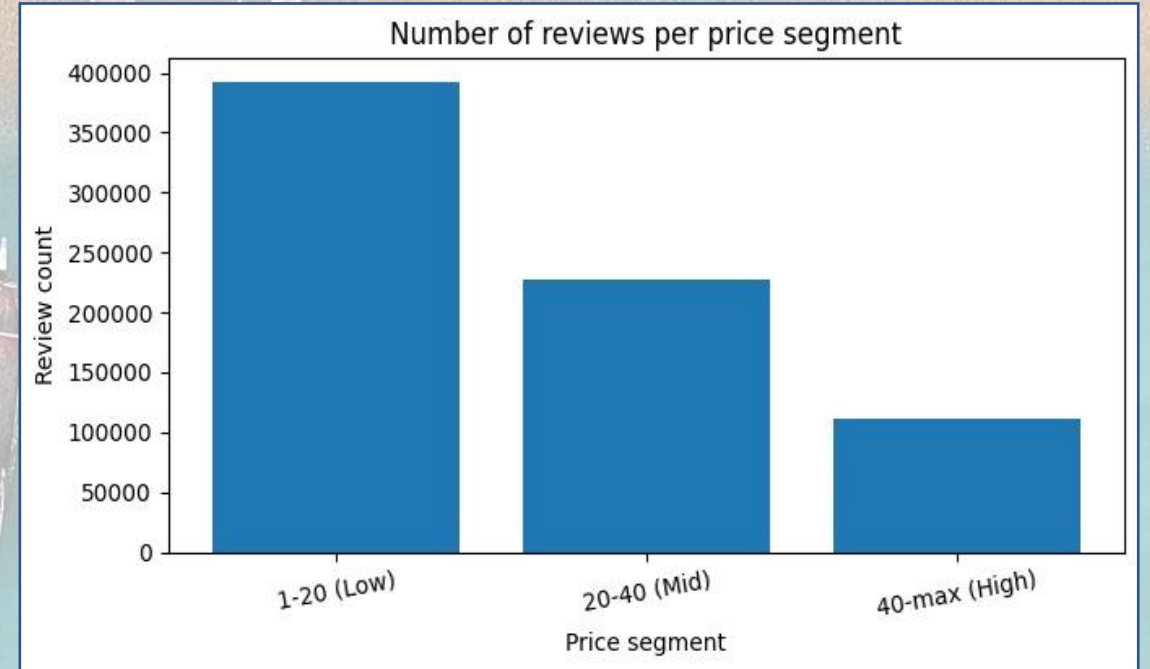
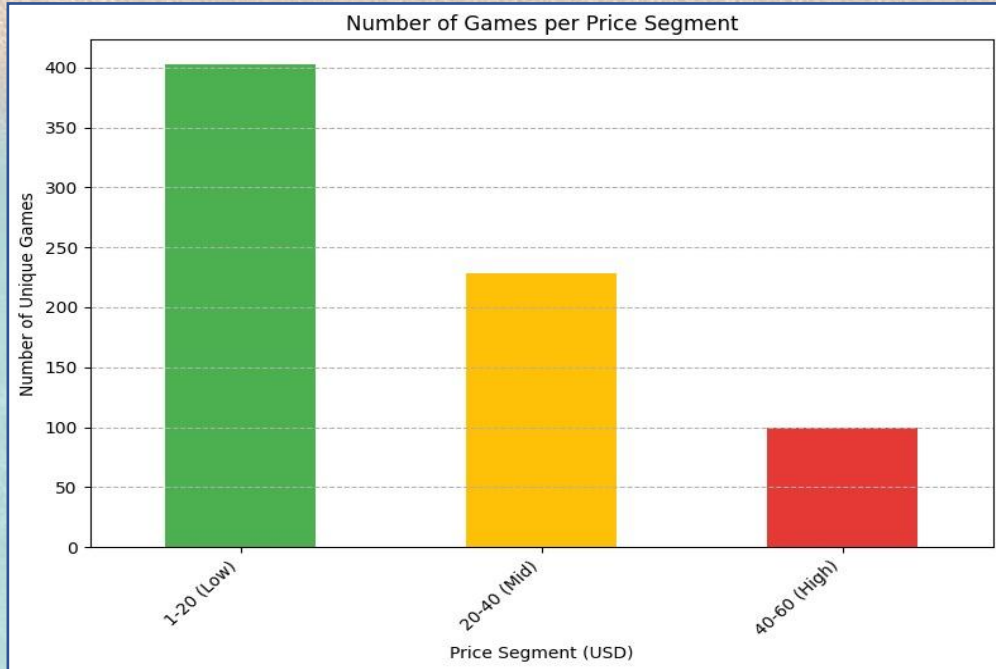
RQ2

Does price act as a signal setting player expectations reflected in review topics?

Analysis Pipeline



Dataset Summary



743

Paid Games

730,945

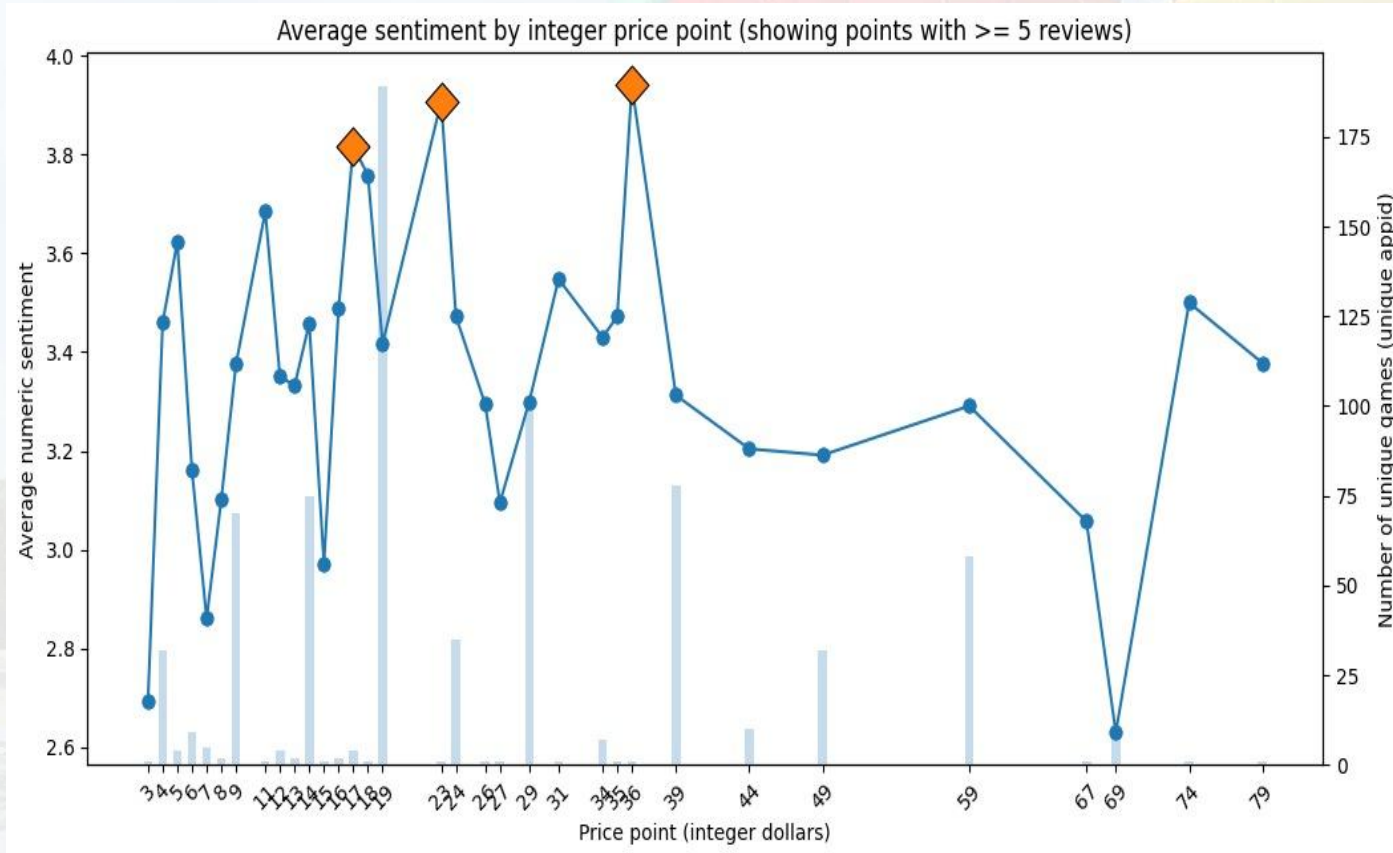
Steam Reviews

~1,000

Reviews Per
Game

Source: SteamSpy API • Filter: Paid games only ($\geq \$1$ USD)

Key Finding #1: Sentiment by Price Segment



Average Sentiment Scores

High-priced (>40 USD): 3.201

Mid-priced (20-40 USD): 3.341 (+4.3%)

Low-priced (1-20 USD): 3.412 (+6.6%)

Lower-priced games receive
slightly more positive sentiment

Regression Analysis: Price vs. Sentiment

Formula: $\text{numeric_sentiment} = \beta_0 + \beta_1 \times \text{price}$

- β (Coefficient): -0.0045 ($p < 0.001$)
- R^2 (Variance Explained): $0.002 \approx 0.2\%$
- Interpretation: For each \$1 increase in price, sentiment decreases by 0.0045
- Critical Finding: Price explains ONLY 0.2% of sentiment variation

Conclusion: Price is NOT a strong predictor of overall review sentiment

Topic Analysis: Categories & Findings

Keyword-based analysis across 730,945 reviews

Top 3 Topics:

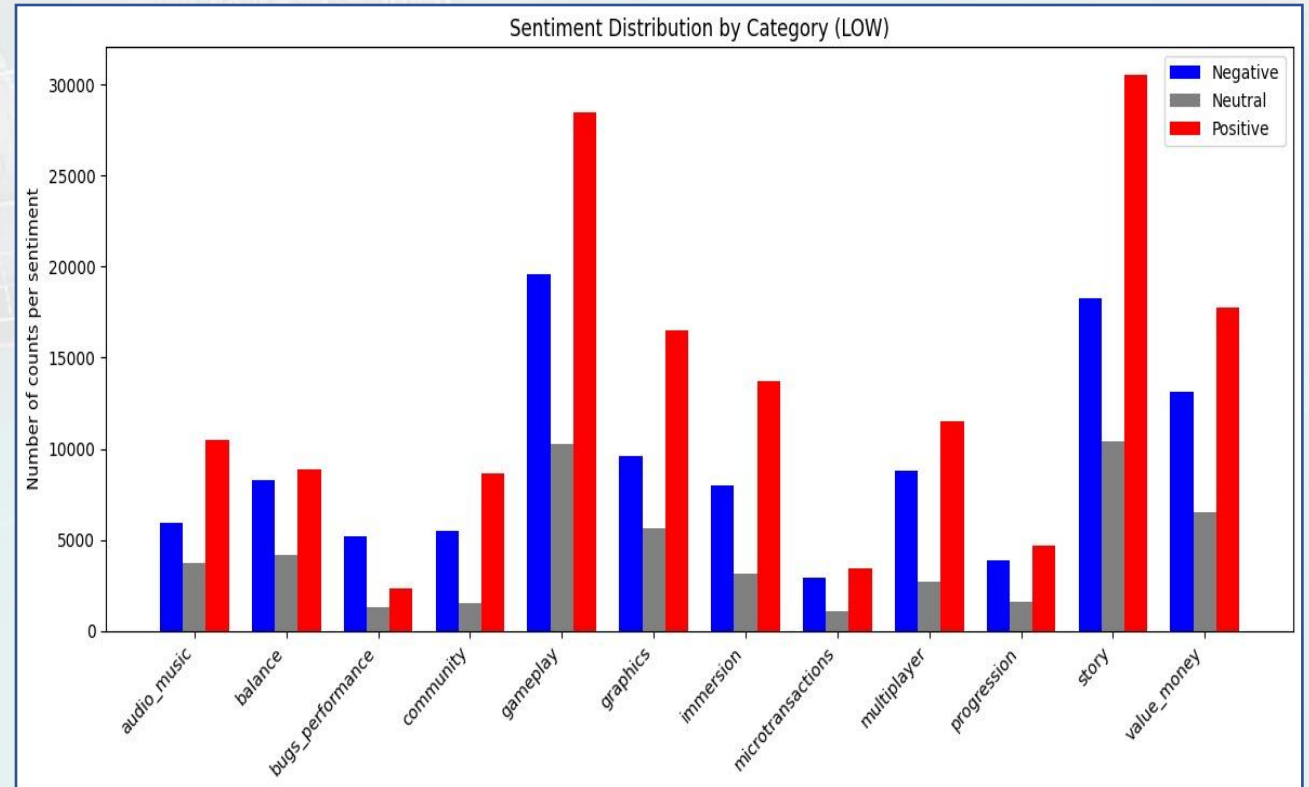
1. **Gameplay** (120,355 mentions)
2. **Story** (119,559 mentions)
3. **Value for Money** (72,386 mentions)



Low-Priced Games (1-20 USD)

Key Characteristics

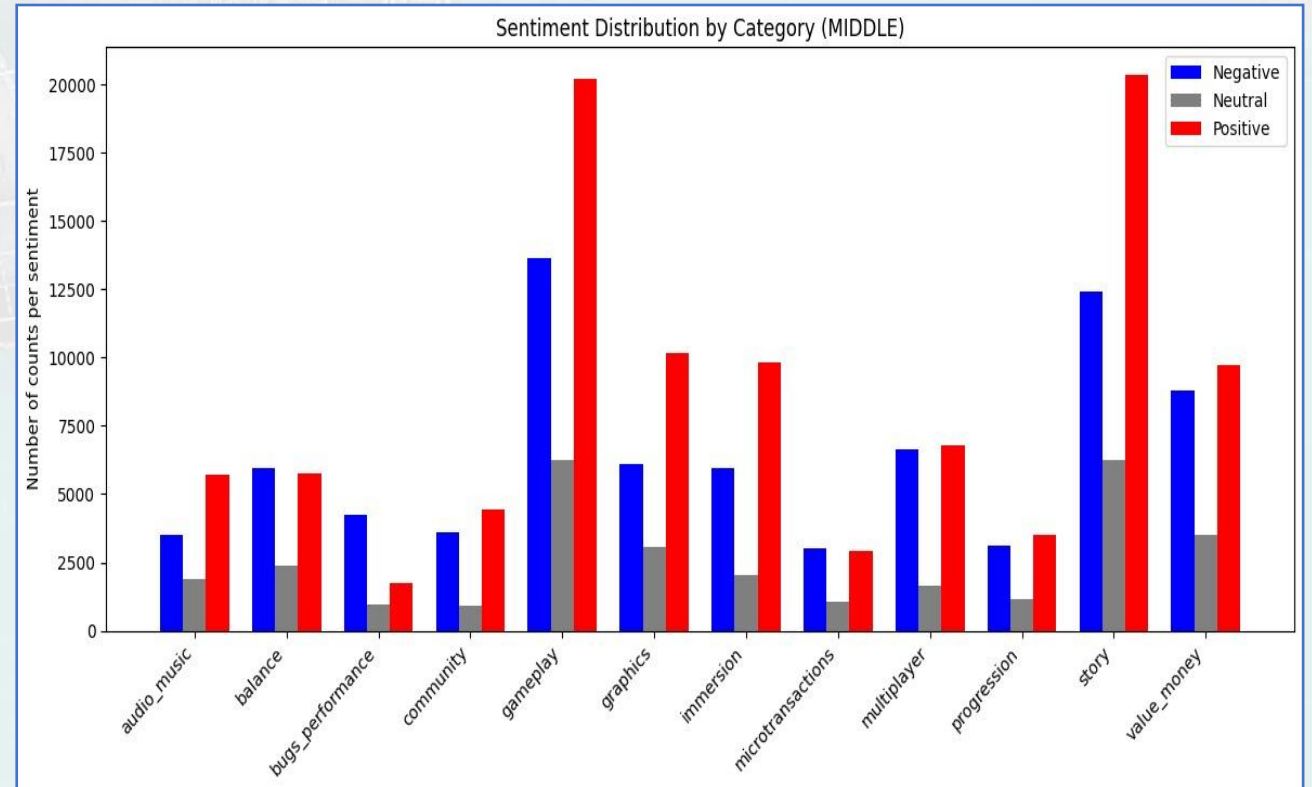
- Highest review volume
- Focus: Value & Story
- Detailed, narrative-focused feedback
- Common complaints: Bugs & Performance
- Overall sentiment: 3.412 (highest)
- Positivity dominates in almost every category.



Mid-Priced Games (20-40 USD)

Key Characteristics

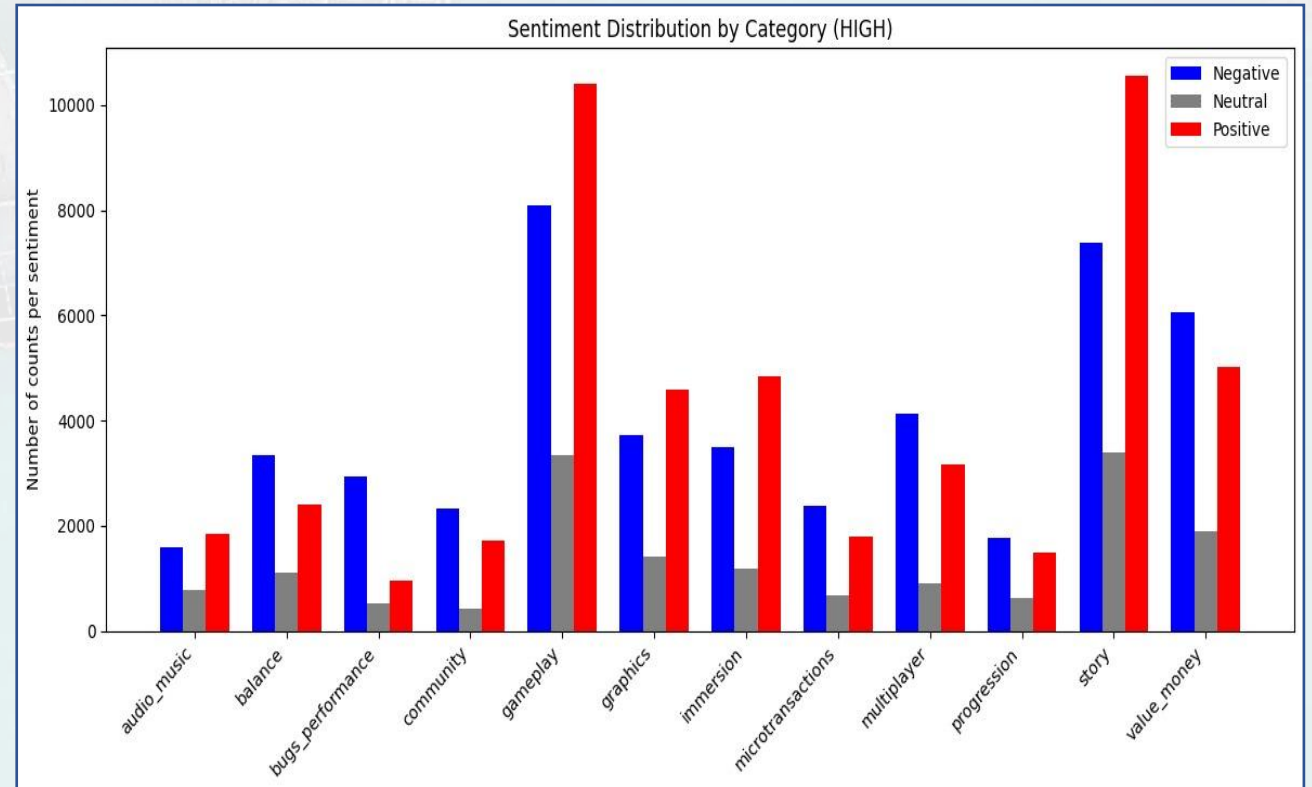
- Moderate review volume
- Focus: **Gameplay & Immersion**
- Balanced feedback on mechanics
- Players expect quality experience
- Overall sentiment: 3.341 (moderate)
- User sentiments starts shifting towards negative.



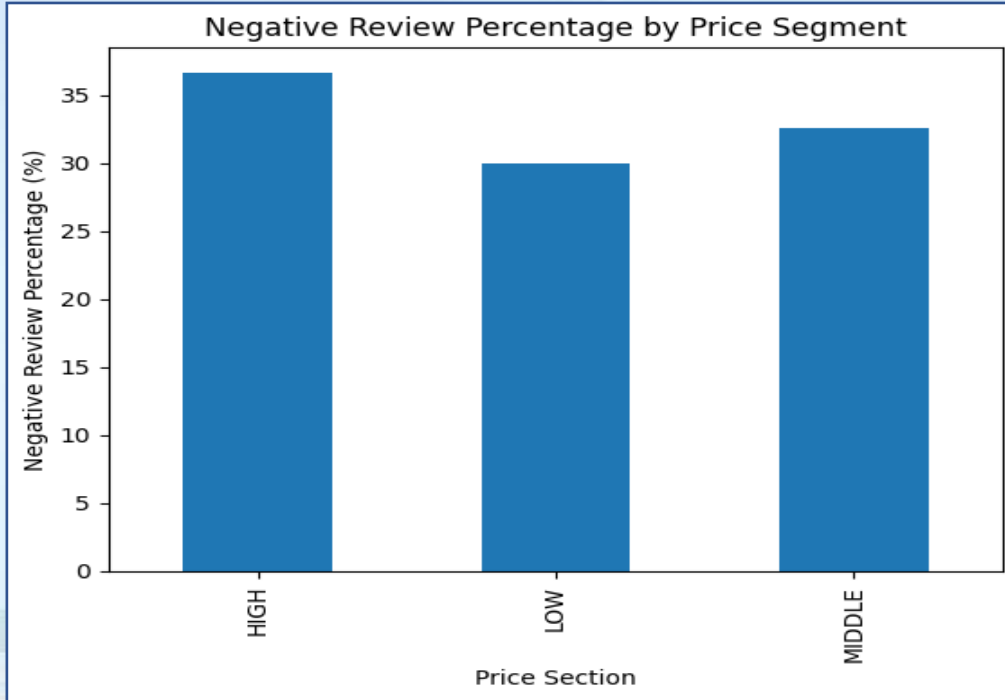
High-Priced Games (>40 USD)

Key Characteristics

- Lowest review volume
- Focus: **Narrow, Focused Feedback**
- More selective player base
- Fewer but more critical reviews
- Overall sentiment: 3.201 (lowest)
- Negative feedbacks dominate in categories such as Balancing, Bugs, Community, Value for Money.



Negativity Dominates in High Priced Games



Price Section	Total Reviews	Negative Reviews	Percentage
High	110842	40646	36.67
Middle	227964	74213	32.56
Low	392139	117568	29.98

Key Findings Summary

Finding 1

- Price has a statistically significant but extremely small negative effect on sentiment.
- In practice, price does NOT meaningfully predict sentiment.

Finding 2

Price acts as a "signal" framing player expectations across different evaluation dimensions

Implications & Impact

For Developers: Pricing signals player expectations; design and market accordingly.

For Publishers: Different price points attract different player priorities.

For Researchers: Price shapes review structure, not sentiment—nuanced market signal.

Limitations & Future Directions

Current Limitations

- Steam-specific data (single platform)
- Manual keyword-based topic analysis

Future Work

- Cross-platform validation (Xbox, PlayStation, mobile)
- Unsupervised topic modeling (LDA)
- Incorporate genre, release date, developer reputation

Conclusion

Price does not predict sentiment, but signals player expectations

Understanding how pricing shapes player discourse—beyond simple sentiment—provides crucial insights for game developers, publishers, and market researchers.

Thank You

Any Questions?